

Nirmal Thomas

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Education

Noida International University
BSc Data Science (CGPA: 8.41)

Noida, UP, India
Sep 2021 – June 2024

Work Experience

Pratham International

Remote – Mumbai, India

AI Engineer

Sep 2024 – Present

- Designed and deployed multiple RAG-based chatbots including:
 - **BaalSakhi** – Deployed on WhatsApp and trained on documents from government departments, UNICEF, WHO, Pratham, pediatricians, and childcare experts; supports mothers in low-resource communities by answering early childhood care queries in regional languages through both text and audio interactions.
 - **ASER Bot** – Developed for India's largest citizen-led education survey providing reliable data on learning outcomes to enable real-time access to ASER data and insights for policymakers and educators using a Hybrid-RAG approach.
 - **Pratham Website Chatbot** – Built a website assistant to help users navigate organizational information efficiently.
- Built speech-to-speech conversational bots using cascaded pipelines, deployed for skilling programs in India and Kenya.
- Built a Voice-based Surveyor for facilitators implementing the Teaching at the Right Level (TaRL) methodology, an AI assistant that identifies evidence-based nudges and personalized pedagogical recommendations from prior facilitator interactions.
- Designed and implemented an in-house translation and transcription pipeline, reducing production costs by over 80% for multilingual video and subtitle generation.

PAiGPT.ai

Noida, UP, India

AI Engineer

Sep 2023 – August 2024

- Built an Answer Engine leveraging LLM agents, advanced prompting, and real-time crawled data for precise, context-aware responses.
- Fine-tuned a pre-trained LLM using Unsloth to accurately generate source citations and improve factual reliability.
- Designed and implemented complete RAG architecture to improve domain adaptation of the proprietary in-house LLM.
- Utilized a vector database to power a scalable recommendation system integrated with the chatbot ecosystem.
- Introduced Hindi text generation using a custom translation pipeline built with an NMT model and CTranslate2, enabling multilingual support.
- Created a data ingestion pipeline using Change Data Capture (CDC) to integrate unstructured data from PDFs and web sources.
- Developed production-grade backend APIs with authentication, rate limiting, and other reliability features.
- Implemented comprehensive security and reliability measures, including prompt injection defense, output guardrails, prompt caching, data encryption, API authentication, and retrieval validation.

Jishu Excellence

Remote - India

Data Science Intern (Part Time)

July 2022 - July 2023

- Assisted in building an Audio Intelligence System that converts call recordings into diarized text with emotion and sentiment analysis to identify prospective customers based on defined metrics.
- Improved overall system reliability by **12%** through evaluation-driven optimization and metric-based feedback loops.
- Enhanced sentiment and emotion detection accuracy by **4%** using integration of state-of-the-art NLP models.
- Developed a Chatbot interface to interact with transcriptions, extract insights, and perform one-click summarization using transformer-based models.

Research Experience

Expedition Tiny Aya

Language Aware Quantization

Hosted by Cohere Labs

March 2026

Summer of Open AI Research

Prompt Optimization for Hallucination Reduction

Hosted by EleutherAI

August 2025

Expedition Aya 2025

Multilingual Speculative Decoding

Hosted by Cohere Labs

April 2025 - May 2025

Publications, Theses and Pre-prints

MRL Workshop Shared Task, EMNLP 2025: Global PIQA: Evaluating Physical Commonsense Reasoning Across 100+ Languages and Cultures [\[Preprint\]](#)

Contributed to the construction of the dataset in Hindi.

Bachelors Dissertation: Decoder Trimming - A Feasible Alternative to Fine-Tuning for Text Classification with Large Language Models

Advised by Dr. Aashima Bangia, Investigated the impact of decoder trimmed LLMs for classification of text.

Awards and Conferences

May 2025: Expedition Aya: Most Promising Award – Explored multilingual speculative decoding to make LLMs more efficient.

February 2025: Best Student Award – Awarded for overall outstanding performance during undergraduate studies.

Skills and Interests

Tools and frameworks: PyTorch, TGI, Transformers, HuggingFace, vLLM, FastAPI, Flask, Lang-Graph, Ctranslate2, Unsloth

Interests: Efficient NLP Methods, Meta Learning,